

ADVANCED LINEAR DATA STRUCTURES

List Types, Stacks & Queues

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Course Outline



1 Advanced List Structures

2 Stacks and Queues



Outline

1 Advanced List Structures

2 Stacks and Queues



Double-linked List: C Structure

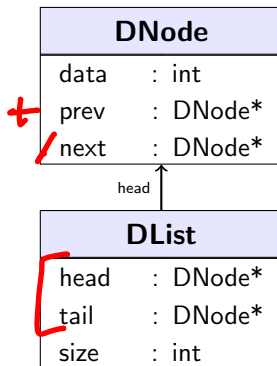
Definition

A **double-linked list** has each **node** pointing to both **next** and **previous** nodes, enabling **bidirectional traversal**.

```
typedef struct DNode {
    int data;
    struct DNode *prev;
    struct DNode *next;
} DNode;

typedef struct {
    DNode *head;
    DNode *tail;
    int size;
} DList;

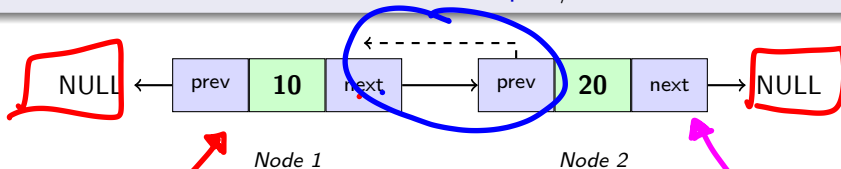
void dl_push_back(DList*, int);
void dl_delete(DList*, DNode*);
DNode *dl_search(DList*, int);
void dl_free(DList*);
```



Double-linked List: Memory Layout

Node diagram

Each node stores three fields; arrows show *prev*/*next* links.



Advantages

- Bidirectional traversal
- $O(1)$ delete given a pointer
- Efficient insert at tail

Disadvantages

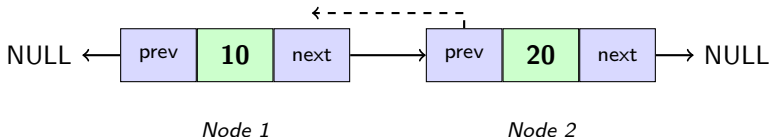
- Extra memory (2 pointers/node)
- More pointer updates per op
- Harder to debug



Double-linked List: Memory Layout

Node diagram

Each node stores three fields; arrows show *prev*/*next* links.



Advantages

- Bidirectional traversal
- $O(1)$ delete given a pointer
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Disadvantages

- Extra memory (2 pointers/node)
- More pointer updates per op
- Harder to debug



Double-linked List: C Operations

```

/* Insert at tail */
void dl_push_back(DList *l,
    int v){
    DNode *n = malloc(sizeof(
        DNode));
    n->data = v; (AA)
    n->next = NULL;
    n->prev = l->tail;
    if (l->tail)
        l->tail->next = n;
    else
        l->head = n;
    l->tail = n;
    l->size++;
}
~

```

```

/* Delete any node
void dl_delete(DList *l, DNode
    *n){
    if (l->prev)
        l->prev->next = n->next;
    else
        l->head = n->next;
    if (n->next)
        n->next->prev = n->prev;
    else
        l->tail = n->prev;
    free(n);
    l->size--;
}

```

Key insight

`dl_delete` is $O(1)$: with a pointer to the node, no search is needed.

Double-linked List: C Operations

```

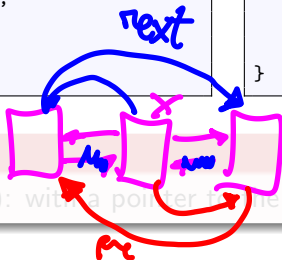
/* Insert at tail */
void dl_push_back(DList *l,
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    DNode *n = malloc(sizeof(
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    n->data = v;
    n->next = NULL;
    n->prev = l->tail;
    if (l->tail)
        l->tail->next = n;
    else
        l->head = n;
    l->tail = n;
    l->size++;
}
~

```

```

/* Delete any node */
void dl_delete(DList *l, DNode
    *n){
    if (n->prev)
        n->prev->next = n->
            next;
    else
        l->head = n->next;
    if (n->next)
        n->next->prev = n->
            prev;
    else
        l->tail = n->prev;
    free(n);
    l->size--;
}

```



Key insight

dl_delete is $O(1)$: with a pointer to the node, no search is needed.

Double-linked List: C Operations

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/* Insert at tail */
void dl_push_back(DList *l,
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    DNode *n = malloc(sizeof(
        DNode));
    n->data = v;
    n->next = NULL;
    n->prev = l->tail;
    if (l->tail)
        l->tail->next = n;
    else
        l->head = n;
    l->tail = n;
    l->size++;
}
~

```

```

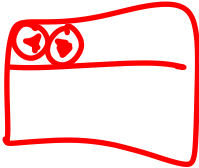
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void dl_delete(DList *l, DNode
    *n){
    if (n->prev)
        n->prev->next = n->
            next;
    else
        l->head = n->next;
    if (n->next)
        n->next->prev = n->
            prev;
    else
        l->tail = n->prev;
    free(n);
    l->size--;
}

```

Key insight

dl_delete is $O(1)$ with a pointer to the node, no search is needed.

Implement **back** / **forward** browser history using a double-linked list.



BrowserHistory	
current	: WebPage*
size	: int
+ visit(url : char*)	
+ go_back()	
+ go_forward()	
+ free_history()	



Why Double-linked List?

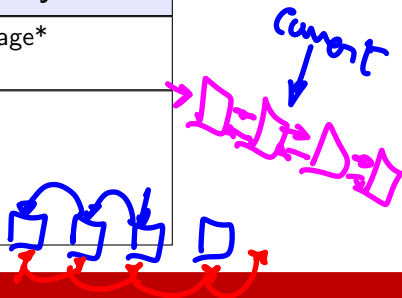
- ~ [udacity.com](https://www.udacity.com) [khanacademy.org](https://www.khanacademy.org) [backblaze.com](https://www.backblaze.com)
 ~ [chatgpt.com](https://www.chatgpt.com) [audible.org](https://www.audible.org)

Study Case: Browser Navigation

Problem

Implement **back** / **forward** browser history using a double-linked list.

BrowserHistory	
current	: WebPage*
size	: int
+ visit(url : char*)	
+ go_back()	
+ go_forward()	
+ free_history()	



Why Double-linked List?

- prev pointer → Back button ($O(1)$)
- next pointer → Forward button ($O(1)$)
- Visiting a new page: truncate forward history from current

Circular List: C Structure

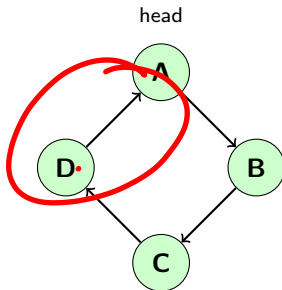
Definition

A **circular list** is a linked list where the **last node's next** points back to the **head**, forming a cycle.

```
typedef struct CNode {
    int data;
    struct CNode *next; /*wraps*/
} CNode;

typedef struct {
    CNode *tail; /*tail->next = head*/
    int size;
} CList;

void cl_insert(CList*, int);
void cl_remove(CList*, int);
CNode *cl_next(CList*); /*rotate*/
```



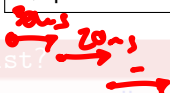
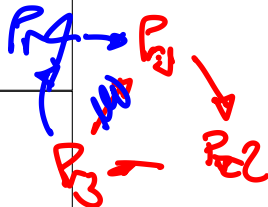
Study Case: Round-Robin Scheduler

Problem

Implement a **round-robin CPU scheduler**: each process receives equal time slices, cycling continuously.

RRScheduler	
current	: Process*
time_quantum	: int
count	: int
+ add_process(pid, burst)	
+ run_quantum()	
+ remove_finished()	
+ print_status()	

Pr. 1
Pr. 2
Pr. 3



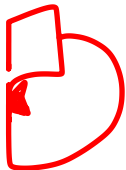
Why Circular List?

The circular structure naturally rotates: after the last process, next leads straight back to the first — no restart logic.

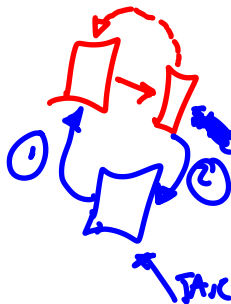
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Why Circular List?

The circular structure **naturally rotates**: after the last process, next leads straight back to the first — no restart logic.

Outline

Pilas $\leftarrow$ Stack $\hookrightarrow$ LIFO
 Last-First

1 Advanced List Structures
 filas $\rightarrow$ filas

Last-In First-Out

2 Stacks and Queues

Queues

filas
 colas

FIFO Last
 First-In First-Out
 first



Stack: LIFO Principle and C Structure

Definition

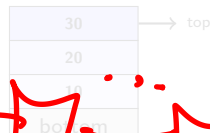
A **stack** follows Last-In-First-Out (LIFO): both insertions (**push**) and deletions (**pop**) occur at the same end, the **top**.

```
typedef struct SNode {
    int data;
    struct SNode *next;
} SNode;
```

```
typedef struct Stack {
    int size;
    SNode *top;
} Stack;
```

```
void push(Stack *s, int v);
int pop (Stack *s);
int peek(const Stack *s);
int is_empty(const Stack *s);
```

insert begin
delete begin



Applications

- Call stack
- Expression evaluation
- Undo systems
- DFS traversal

insert end
delete end



Stack: LIFO Principle and C Structure

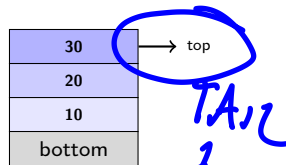
Definition

A **stack** follows **Last-In-First-Out (LIFO)**: both insertions (push) and deletions (pop) occur at the same end, the **top**.

```
typedef struct SNode {
    int data;
    struct SNode *next;
} SNode;

typedef struct {
    SNode *top;
    int size;
} Stack;

void push(Stack *s, int v);
int pop(Stack *s);
int peek(const Stack *s);
int is_empty(const Stack *s);
```



Applications

- Call stack
- Expression evaluation
- Undo systems
- DFS traversal

Push ← Insertion
Pop ← delete-er

Stack: LIFO Principle and C Structure

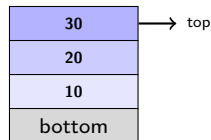
Definition

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```
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    int data;
    struct SNode *next;
} SNode;

typedef struct {
    SNode *top;
    int size;
} Stack;

void push(Stack *s, int v);
int pop (Stack *s);
int peek(const Stack *s);
int is_empty(const Stack *s);
```



Applications

- Call stack
- Expression evaluation
- Undo systems
- DFS traversal



Stack: C Implementation

```

void push(Stack *s, int v) {
    (SNode *n = malloc(sizeof(
        SNode)));
    n->data = v;
    n->next = s->top;
    s->top = n;
    s->size++;
}

int is_empty(const Stack *s) {
    return s->top == NULL;
}

void free_stack(Stack *s) {
    while (!is_empty(s))
        pop(s);
}
  
```

Handwritten notes: A red circle around the opening parenthesis of the first function call, and a red arrow pointing to the `s->top` variable with the word "HEAD" written next to it.

```

int pop(Stack *s) {
    if (!s->top) return -1;
    SNode *tmp = s->top;
    int val = tmp->data;
    s->top = tmp->next;
    free(tmp);
    s->size--;
    return val;
}

int peek(const Stack *s) {
    if (!s->top) return -1;
    return s->top->data;
}
  
```

Handwritten notes: A red circle around the `-1` return value in the `pop` function, and a red underline under the `tmp->next` expression.

Memory safety

Always call `free_stack()` before the stack goes out of scope to avoid memory leaks.

Stack: C Implementation

```

void push(Stack *s, int v) {
    SNode *n = malloc(sizeof(
        SNode));
    n->data = v;
    n->next = s->top;
    s->top = n;
    s->size++;
}

int is_empty(const Stack *s) {
    return s->top == NULL;
}

void free_stack(Stack *s) {
    while (!is_empty(s)) {
        pop(s);
    }
}

```

```

int pop(Stack *s) {
    if (!s->top) return -1;
    SNode *tmp = s->top;
    int val = tmp->data;
    s->top = tmp->next;
    free(tmp);
    s->size--;
    return val;
}

int peek(const Stack *s) {
    if (!s->top) return -1;
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}

```

Memory safety

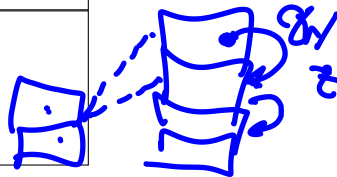
Always call `free_stack()` before the stack goes out of scope to avoid memory leaks.

Study Case: Undo / Redo System

Problem

Implement **undo** / **redo** for a text editor using two stacks.

UndoManager
undo_stack : Stack redo_stack : Stack
+ record(action : EditAction) + undo() + redo() + clear()



Logic

- **Record:** push to undo_stack, clear redo_stack
- **Undo:** pop from undo_stack, push to redo_stack
- **Redo:** pop from redo_stack, push to undo_stack

Queue: FIFO Principle and C Structure

Definition

A **queue** follows First-In-First-Out (FIFO): elements are enqueued at the rear and dequeued from the front.

```

typedef struct QNode {
    int data;
    struct QNode *next;
} QNode;

typedef struct {
    QNode *front;
    QNode *rear;
    int size;
} Queue;

void enqueue(Queue *q, int v);
int dequeue(Queue *q);
int q_front(const Queue *q);
int q_empty(const Queue *q);
  
```

Handwritten annotations:

- remove* (above `QNode`)
- insertion* (next to `next`)
- enqueue* (next to `enqueue`)
- insert-begin* (next to `enqueue`)



Applications

- CPU scheduling
- I/O buffers
- BFS traversal
- Service queues

Handwritten annotations:

- dequeue* (next to the first application)
- delete-end* (next to the second application)

Queue: FIFO Principle and C Structure

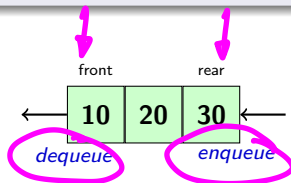
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Queue: FIFO Principle and C Structure

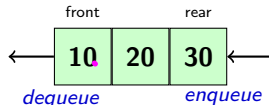
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```



Applications

- CPU scheduling
- I/O buffers
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Queue: C Implementation

```

void enqueue(Queue *q, int v)
{
    QNode *n = malloc(sizeof(
        QNode));
    n->data = v;
    n->next = NULL;
    if (q->rear)
        q->rear->next = n;
    else
        q->front = n;
    q->rear = n;
    q->size++;
}

```

```

int dequeue(Queue *q) {
    if (!q->front)
        return -1;
    QNode *tmp = q->front;
    int val = tmp->data;
    q->front = tmp->next;
    if (!q->front)
        q->rear = NULL;
    free(tmp);
    q->size--;
    return val;
}
~

```

Key difference from Stack

Stack maintains **one pointer** (top). Queue maintains **two**: front and rear.



Queue: C Implementation

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}
```

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    int val = tmp->data;
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~
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Key difference from Stack

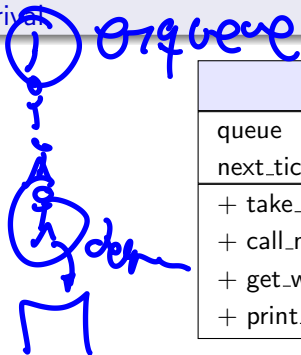
Stack maintains one pointer (top). Queue maintains two: front and rear.



Study Case: Bank Queue System

Problem

Implement a **bank teller queue**: customers arrive and are served **in order of arrival**.



BankQueue	
queue	: Queue
next_ticket	: int
+ take_ticket(name : char*)	
+ call_next()	
+ get_wait_count()	
+ print_status()	

Why Queue?

FIFO guarantees **fair service**: the first customer to arrive is the first to be served. No customer can be skipped.

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2 Stacks and Queues



Complexity Summary

Operation	Time	Notes
Stack push	$O(1)$	Prepend to top
Stack pop	$O(1)$	Remove from top
Stack peek	$O(1)$	Read without removal
Queue enqueue	$O(1)$	Append to rear
Queue dequeue	$O(1)$	Remove from front
DList insert at ends	$O(1)$	Requires tail pointer
DList dl_delete	$O(1)$	Pointer to node given
DList dl_search	$O(n)$	Sequential scan

Space complexity

All linked implementations: $O(n)$ for n elements.

Overhead: 1 pointer / node (singly), 2 pointers / node (doubly).



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

